

4MA1_2H_ch2_equations_formulae_identities

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1. 4MA1/2H_Winter_2024_Q23

Show that $\frac{16x^2 - 36}{x - 7} \div \frac{2x^2 + 7x + 6}{x^2 - 5x - 14} - (7 + 8x) = n$

where n is an integer to be found.

Show clear algebraic working.

4 marks

2. 4MA1/2H_Winter_2024_Q22

Solve the simultaneous equations

$$x^2 + y^2 + y = 3$$

$$x + 2 = y$$

Show clear algebraic working.

5 marks

3. 4MA1/2H_Winter_2024_Q20

Solve the inequality $10x^2 + 11x - 21 < 0$

Show clear algebraic working.

3 marks

4. 4MA1/2H_Winter_2024_Q12

(a) Solve $\frac{3x+2}{5} - \frac{2x+1}{3} = x$

Show clear algebraic working.

$x = \dots\dots\dots$
(3)

(b) Make c the subject of the formula $f = \sqrt{\frac{a+bc}{c-d}}$

$\dots\dots\dots$
(4)

5. 4MA1/2HR_Summer_2024_Q26

Write $4 - \left[(3x - 5) \div \frac{3x^2 + x - 10}{4x - 1} \right]$ as a single fraction in its simplest form.

(4)

6. 4MA1/2HR_Summer_2024_Q22

Solve the simultaneous equations

$$\begin{aligned} x^2 + y^2 &= y + 11 \\ y &= 3x - 1 \end{aligned}$$

Show clear algebraic working.

(5)

7. 4MA1/2HR_Summer_2024_Q20

Solve $6x^2 - 7x - 20 > 0$

Show clear algebraic working.

(4)

8. 4MA1/2HR_Summer_2024_Q6

- (a) Solve $\frac{2f}{3} = 4f - 17$
Show clear algebraic working.

$f = \dots\dots\dots$
(3)

- (b) Simplify $(e + 12)^0$ where $e > 0$
(1)

- (c) Simplify fully $\frac{12a^4h^6}{4ah^2}$
(2)

- (d) Factorise fully $20x^5y + 12x^3y^4$
(2)

9. 4MA1/2H_Summer_2024_Q23

Simplify $\frac{30 \times 25^{2x+7}}{\sqrt{180} \times (\sqrt{5})^{4x+9}}$

Give your answer in the form 5^w where w is an expression in terms of x
Show each stage of your working clearly.
(3)

10. 4MA1/2H_Summer_2024_Q20

Given that $k = x - y$ and $x = \frac{1}{4y}$

express $\frac{5k}{x+2}$ in the form $\frac{a-by^2}{c+dy}$ where a , b , c and d are integers.

(3)

11. 4MA1/2H_Summer_2024_Q17

Q is directly proportional to the square root of d

$Q = 4.5$ when $d = 324$

Find a formula for Q in terms of d

(3)

12. 4MA1/2H_Summer_2024_Q15

(a) Make g the subject of $e = \sqrt{\frac{7g+5}{11+2g}}$ (4)

(b) Solve the inequality $3y^2 + 4y - 32 > 0$
Show your working clearly. (3)

13. 4MA1/2H_Summer_2024_Q14

(a) Show that $3y(2y+5)(y+7)$ can be written in the form $ay^3 + by^2 + cy$ where a, b and c are integers. (3)

(b) Solve $\frac{2x+3}{5} + \frac{6x-5}{4} = \frac{163}{100}$

Show clear algebraic working.

$x =$ (4)

14. 4MA1/2H_Summer_2024_Q11

(i) Factorise $x^2 + 9x - 22$ (2)

(ii) Hence, solve $x^2 + 9x - 22 = 0$ (1)

15. 4MA1/2H_Summer_2024_Q10

Solve the simultaneous equations

$$\begin{aligned} 6x + 4y &= 1 \\ 3x + 5y &= 8 \end{aligned}$$

Show clear algebraic working.

$x = \dots\dots\dots$
 $y = \dots\dots\dots$
(3)

Write

$$\frac{4x^2 - 17x - 15}{2x - 1} \times \frac{2x^2 - 7x + 3}{x^2 - 25} + (29 - 4x)$$

as a single fraction in its simplest form.

.....

4 marks

Solve the simultaneous equations

$$\begin{aligned}2y^2 + x^2 &= -6x + 42 \\ 2x + y &= -3\end{aligned}$$

Show clear algebraic working.

.....

5 marks

18. 4MA1/2HR_Winter_2023_Q19

Prove algebraically that, for any three consecutive even numbers,

the sum of the squares of the smallest even number and the largest even number is
8 more than twice the square of the middle even number.

3 marks

(a) Simplify $\frac{2}{y^6}$

(1)

(b) Simplify fully $(16a^4)^{\frac{3}{4}}$

(2)

(c) Expand and simplify $5x(3x + 4)(2x - 1)$

(3)

Find the values of n such that

$$\frac{10^{4n} \times 2^{3(n^2-5n)} \times 5^{2(1-2n)}}{20^2} = 1$$

Show clear algebraic working.

.....

5 marks

Solve the inequality $6x^2 + 37x \leq 35$
Show clear algebraic working.

.....

3 marks

Solve the simultaneous equations

$$\begin{aligned}y &= 7 - 2x \\ x^2 + y^2 &= 34\end{aligned}$$

Show clear algebraic working.

.....

5 marks

Given that $8\sqrt{m} + \sqrt{49m} - \sqrt{9m} = k\sqrt{m}$

where k is an integer and m is a prime number,

(a) work out the value of k

$$k = \dots\dots\dots (1)$$

(b) Show that $\frac{5 - \sqrt{18}}{1 - \sqrt{2}}$ can be written in the form $a + b\sqrt{2}$

where a and b are integers.

Show each stage of your working clearly.

(3)

Use algebra to show that $0.4\dot{3}\dot{8} = \frac{217}{495}$

2 marks

Solve $\frac{x+3}{4} - \frac{7-x}{5} = 4.3$

Show clear algebraic working.

$x =$

3 marks

- (a) Write down the value of $(m + 2)^0$ where m is a positive integer.

.....
(1)

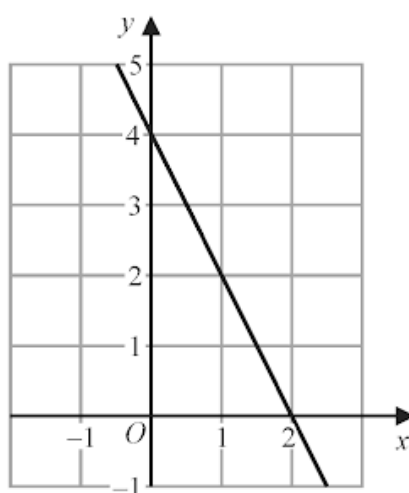
- (b) Simplify $(3a^2b^4)^5$

.....
(2)

- (c) Factorise fully $14x^2y^4 + 21x^3y^2$

.....
(2)

The diagram shows a straight line drawn on a grid.



- (d) Write down an equation of the line.

.....
(2)

Solve the simultaneous equations

$$\begin{aligned}5x + 4y &= -2 \\ 2x - y &= 4.4\end{aligned}$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

3 marks

The diagram shows rectangle $ABCD$

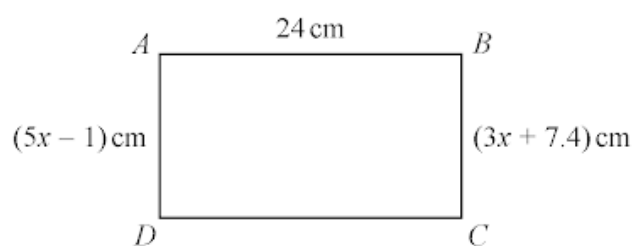


Diagram **NOT**
accurately drawn

Work out the perimeter of the rectangle.
Show your working clearly.

..... cm

4 marks

Work out the coordinates of the points of intersection of

$$y - 2x = 1 \quad \text{and} \quad y^2 + xy = 7$$

Show clear algebraic working.

(.....,.....)

(.....,.....)

5 marks

Make x the subject of $y = \sqrt[3]{\frac{6+5x}{x+4}}$

.....

4 marks

31. 4MA1/2HR_Summer_2023_Q11

Express $\left(\frac{m^6 k^{10}}{25}\right)^{\frac{3}{2}}$ in the form $\frac{m^a k^b}{c}$ where a , b and c are integers to be found.

.....
2 marks

32. 4MA1/2HR_Summer_2023_Q7

(a) Simplify $(4^{-2})^0$

.....
(1)

$$3^{-14} \times 3^8 = 3^m$$

(b) Find the value of m

$m =$
(1)

$$P = m^2 - 4c$$

(a) Work out the value of P when $m = -5$ and $c = 3$

$$P = \dots\dots\dots$$

(2)

(b) Expand and simplify $(x + 5)(x - 7)$

$$\dots\dots\dots$$

(2)

Solve the simultaneous equations

$$\begin{aligned}2x^2 + 3y^2 &= 11 \\ x &= 3y - 1\end{aligned}$$

Show clear algebraic working.

.....

5 marks

Express $3x^2 - 6x + 5$ in the form $a(x - b)^2 + c$

.....

3 marks

(a) Expand and simplify $(x + 6)(3x - 2)(x + 6)$

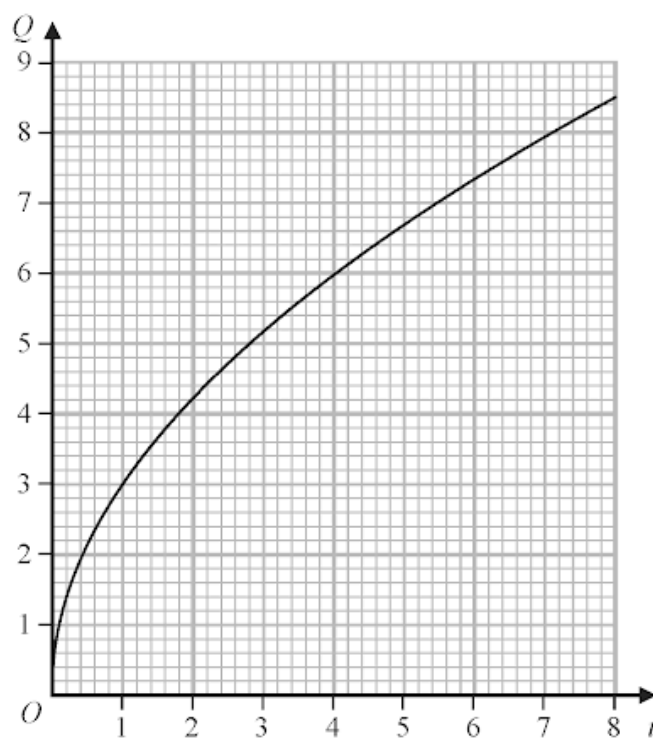
.....
(3)

(b) Make e the subject of $w = \sqrt{\frac{e + g}{ef - d}}$

.....
(4)

Q is directly proportional to $\sqrt{t}$

The graph shows the relationship between Q and t for $0 < t < 8$



(a) Find a formula for Q in terms of t

.....
(3)

Q is increased by 20%

(b) Find the percentage increase in t

..... %
(2)

(a) Simplify fully $(32a^{15})^{\frac{3}{5}}$

.....
(2)

(b) Express $\left(\frac{1}{10x}\right)^{-3}$ in the form px^n where p and n are integers.

.....
(2)

(c) Solve $\frac{1-2y}{3} = \frac{4}{5} - \frac{2y-1}{2}$

Show clear algebraic working.

$y =$
(3)

y is inversely proportional to $\sqrt{x}$
 x is directly proportional to T^3

Given that $y = 8$ when $T = 25$

find the exact value of T when $y = 27$

$T = \dots\dots\dots$

4 marks

Make t the subject of $n^2 = \frac{4d+t^3}{t^3}$

.....

4 marks

- (a) Write down the value of x^0

.....
(1)

Given that $2^{-3} \times 2^9 = 2^n$

- (b) find the value of n

$n =$
(1)

Given that $\frac{7^{206} \times 7^m}{7^{214}} = 7^{-3}$

- (c) find the value of m

$m =$
(2)

(a) Solve $4y + 5 > 12$

.....
(2)

(b) Solve $6x - 5 = \frac{4x - 7}{2}$

Show clear algebraic working.

$x =$
(3)

Here is a rectangle.

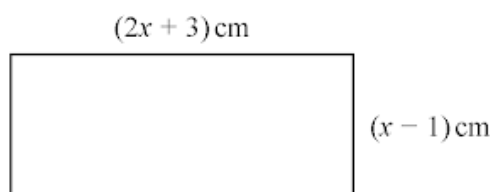


Diagram **NOT**
accurately drawn

Given that the area of the rectangle is less than 75 cm^2

find the range of possible values of x

.....
5 marks

Prove that when the sum of the squares of any two consecutive odd numbers is divided by 8, the remainder is always 2
Show clear algebraic working.

3 marks

Solve the simultaneous equations

$$3x + 5y = 3.1$$

$$6x + 3y = 3.75$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

3 marks

(a) Expand and simplify $(y + 4)(2 - y)$

.....
(2)

(b) Factorise fully $15b^5c - 35b^3c^9$

.....
(2)

Express $\left(\frac{20}{x^2 - 36} - \frac{2}{x - 6}\right) \times \frac{1}{4 - x}$ as a single fraction in its simplest form.

.....

3 marks

Solve the simultaneous equations

$$\begin{aligned}x - 2y &= 3 \\ x^2 - y^2 + 2x &= 10\end{aligned}$$

Show clear algebraic working.

.....

5 marks

49. 4MA1/2HR_Summer_2022_Q17

Use algebra to show that $0.\dot{3}4\dot{5} = \frac{19}{55}$

2 marks

$$\frac{2^k}{4^n} = 2^x$$

Find an expression for x in terms of k and n

$x =$

2 marks

(a) Solve $p = \frac{3p-5}{10}$

Show clear algebraic working.

$$p = \dots\dots\dots$$

(3)

(b) Simplify a^0 where $a > 0$

$$\dots\dots\dots$$

(1)

(c) Simplify fully $\frac{3xy^3}{6x^2y}$

$$\dots\dots\dots$$

(2)

(d) Factorise fully $10c^3d^2 + 15cd^4$

$$\dots\dots\dots$$

(2)

$$\frac{18 \times (\sqrt{27})^{4n+6}}{6 \times 9^{2n+8}} = 3^x$$

Express x in terms of n

Show your working clearly and simplify your expression.

$x = \dots\dots\dots$

3 marks

$a = \frac{14}{3x - 7} \qquad x = \frac{7}{4y - 3}$

Express a in the form $\frac{py + q}{ry + s}$ where p, q, r and s are integers.

Give your answer in its simplest form.

$a = \dots\dots\dots$

3 marks

54. 4MA1/2H_Summer_2022_Q17

M varies directly as the cube of h

$M = 4$ when $h = 0.5$

Find the value of h when $M = 500$

.....
4 marks

(a) Solve $\frac{4x+5}{3} - \frac{3-2x}{2} = 13$

Show clear algebraic working.

$$x = \dots\dots\dots$$

(4)

(b) Solve the inequality $2y^2 - 7y - 30 \leq 0$
Show your working clearly.

$$\dots\dots\dots$$

(3)

- (a) Expand and simplify $(5 - x)(2x + 3)(x + 4)$
Show your working clearly.

.....
(3)

- (b) Make c the subject of $g = \frac{c + 3}{4 + c} - 7$

.....
(4)

57. 4MA1/2H_Summer_2022_Q11

(i) Factorise $x^2 + 5x - 24$

.....
(2)

(ii) Hence, solve $x^2 + 5x - 24 = 0$

.....
(1)

Solve the simultaneous equations

$$7x + 3y = 3$$

$$3x - y = 7$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

3 marks

Solve the equation

$$\frac{5}{x+2} + \frac{3}{x^2+2x} = 2$$

Show clear algebraic working.

.....
(Total for Question is 5 marks)

60. 4MA1/2HR_Winter_2021_Q15

Prove algebraically that the product of any two odd numbers is always an odd number.

(Total for Question is 4 marks)

Solve the simultaneous equations $2x + 7y = 17$
 $5x + 3y = -1$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question is 4 marks)

Solve $x^2 - 21x + 20 = 0$
Show your working clearly.

.....
(Total for Question is 3 marks)

(a) Simplify $(3k^2)^4$

(2)

(b) Simplify $(21m^4n) \div (3n^{-5})$

(2)

$$w = 5y^2 - y^3$$

(a) Work out the value of w when $y = -2$

$$w = \dots\dots\dots$$

(2)

(b) Factorise fully $8p^2 - 2p$

$$\dots\dots\dots$$

(2)

(c) Expand $4t(3t - 2)$

$$\dots\dots\dots$$

(2)

(d) Expand and simplify $(5x - 2)(x + 4)$

$$\dots\dots\dots$$

(2)

Write $\frac{25x^2 - 64}{5x^2 - 13x - 6} \times \frac{x^2 - 8x + 15}{5x + 8} - (x - 7)$

as a single fraction in its simplest form.
Show clear algebraic working.

(a) Solve $\frac{9a-7}{5} - \frac{3a-7}{4} = 4.55$

Show clear algebraic working.

$$a = \dots\dots\dots$$

(3)

(b) Make c the subject of the formula $p = \sqrt{\frac{ac+8}{3+c}}$

$$\dots\dots\dots$$

(4)

Simplify fully $\left(\frac{9t^4w^9}{18t^6w^{10}}\right)^{-2}$

.....
(Total for Question is 3 marks)

(a) Expand and simplify $4x(2x + 5) - 3x(2x - 3)$

.....
(2)

Given that $\frac{y^5 \times y^n}{y^6} = y^{13}$

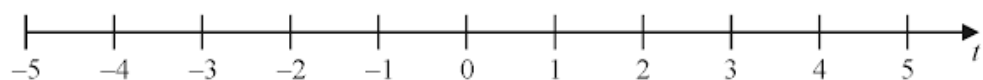
(b) work out the value of n .

$n =$
(2)

(c) (i) Solve the inequality $7t - 8 < 2t + 7$

.....
(2)

(ii) On the number line below, represent the solution set of the inequality solved in part (c)(i)



(1)

Solve the simultaneous equations

$$\begin{aligned}y &= 3 - 2x \\ x^2 + y^2 &= 18\end{aligned}$$

Show clear algebraic working.

.....

5 marks

70. 4MA1/2H_Summer_2021_Q14

Simplify fully $\left(\frac{9x^4}{16y^{10}}\right)^{-\frac{1}{2}}$

.....
3 marks

(i) Factorise $x^2 + 2x - 24$

.....
(2)

(ii) Hence solve $x^2 + 2x - 24 = 0$

.....
(1)

Solve the simultaneous equations

$$5a + 2c = 10$$

$$2a - 4c = 7$$

Show clear algebraic working.

$$a = \dots\dots\dots$$

$$c = \dots\dots\dots$$

3 marks

(a) Simplify fully $\frac{10x^2 + 23x + 12}{4x^2 - 9}$

$$2^{2y} \times 2^{3y+2} = \frac{8^{5y}}{4^n}$$

.....
(3)

- (b) Find an expression for n in terms of y .
Show clear algebraic working and simplify your expression.

.....
(4)

(a) Solve $\frac{4-3x}{5} - \frac{3x-5}{2} = -3$

Show clear algebraic working.

$x = \dots\dots\dots$
(3)

(b) Solve the inequality $5y^2 - 17y \leq 40$

$\dots\dots\dots$
(3)
